Project Design Phase-I
Solution Architecture

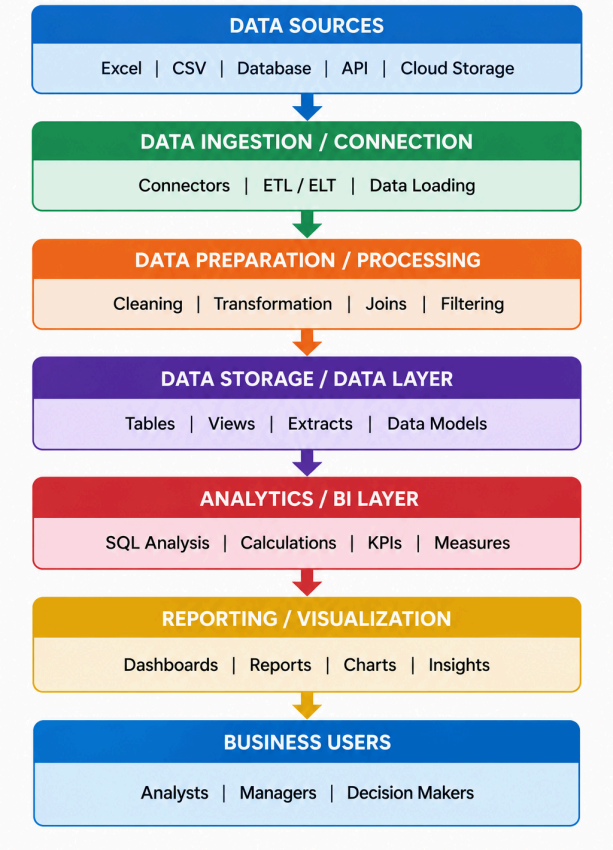
Date	19 September 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Project - xxx
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used

Proposed Solution Architecture

Flow: Excel workbook → Tableau data source and relationships → data validation/calculated fields → worksheets → interactive dashboards → Tableau Story → optional Tableau Public publication.

Use race_results as the central result-level sheet. Relate circuits using circuit_id; use championship_standings for season standings; use qualifying_results and lap_intelligence for qualifying/lap analysis. Validate cardinality and avoid row multiplication before combining sheets.

Snowflake is an optional deployment path, not assumed to exist. The delivered solution is Tableau-oriented unless the team independently provisions Snowflake.

Component	Input / output	Responsibility
Source workbook	Five sheets → Tableau data source	Keep a read-only source copy.
Data validation	Data source → verified fields	Check nulls, data types, key grain and relationship behavior.
Analysis layer	Calculated fields and worksheets	Build documented KPIs and visualizations.
Dashboard layer	Worksheets → dashboards	Coordinate filters, legends, tooltips and layout.
Story layer	Dashboards → story points	Communicate evidence in a clear sequence.
Publication	Workbook → Tableau Public if approved	Insert real public URL only after publishing.